Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	28 October 2026
Team ID	
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	4 Marks

Functional Requirements:
Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Ingest and integrate 5,000 anonymized patient records (smart_healthcare_dataset.csv) comprising clinical vitals (blood pressure, glucose, cholesterol, BMI) and lifestyle habits (exercise, smoking, alcohol).	High
2	Data Processing	Clean missing records, normalize continuous clinical parameters, and perform feature engineering to categorize Age Brackets, WHO BMI categories, and calculate the Dynamic Risk Tier Index (High, Moderate, Low Risk).	High
3	Business Analysis	Compute population-level KPIs including cohort average blood pressure (134.4 mmHg), average glucose (134.5 mg/dL), average BMI (25.0 kg/m^2), and multi-variable correlations between lifestyle factors and chronic conditions.	High
4	Business Insights & Analytical Outputs	Render 9 analytical worksheets, 2 interactive dashboards (Executive Population Overview & Clinical Risk Drivers), and a guided 4-scene Tableau Executive Story container.	High

5	User Interaction & Analysis	Provide dynamic interactive filters enabling clinicians and executives to slice data by Age Brackets, WHO BMI categories, Risk Tiers, and Disease Co-occurrence heatmaps.	Medium
6	Data Update & Reporting	Embed Tableau analytical views into a Python Flask web application (app.py) via Tableau JS API v2, hosting live analytics at http://127.0.0.1:5001.	High

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Intuitive UI/UX design with responsive web container, clean typography, executive KPI stat cards, and seamless 4-scene narrative story navigation.
NFR-2	**Performance & Latency**	Interactive filter response times under 1.5 seconds and initial Tableau dashboard embedded frame load time within 3.0 seconds over Flask WSGI server.
NFR-3	**Scalability**	Flask web architecture and Tableau JS API integration capable of supporting expanding dataset sizes (up to 100,000+ patient records) without performance degradation.
NFR-4	**Reliability & Availability**	99.9% local server availability on port 5001 with robust Flask error handling and automatic debug reloading for seamless deployment.
NFR-5	**Security & Data Privacy**	Compliance with HIPAA data handling principles by using anonymized patient IDs and preventing unauthorized raw data mutations through read-only web embedding.
NFR-6	**Compatibility**	Cross-browser compatibility across Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari across desktop and tablet screen resolutions.

Figure 1: Health360 Live Web Solution Architecture & Embedded Tableau Interface

